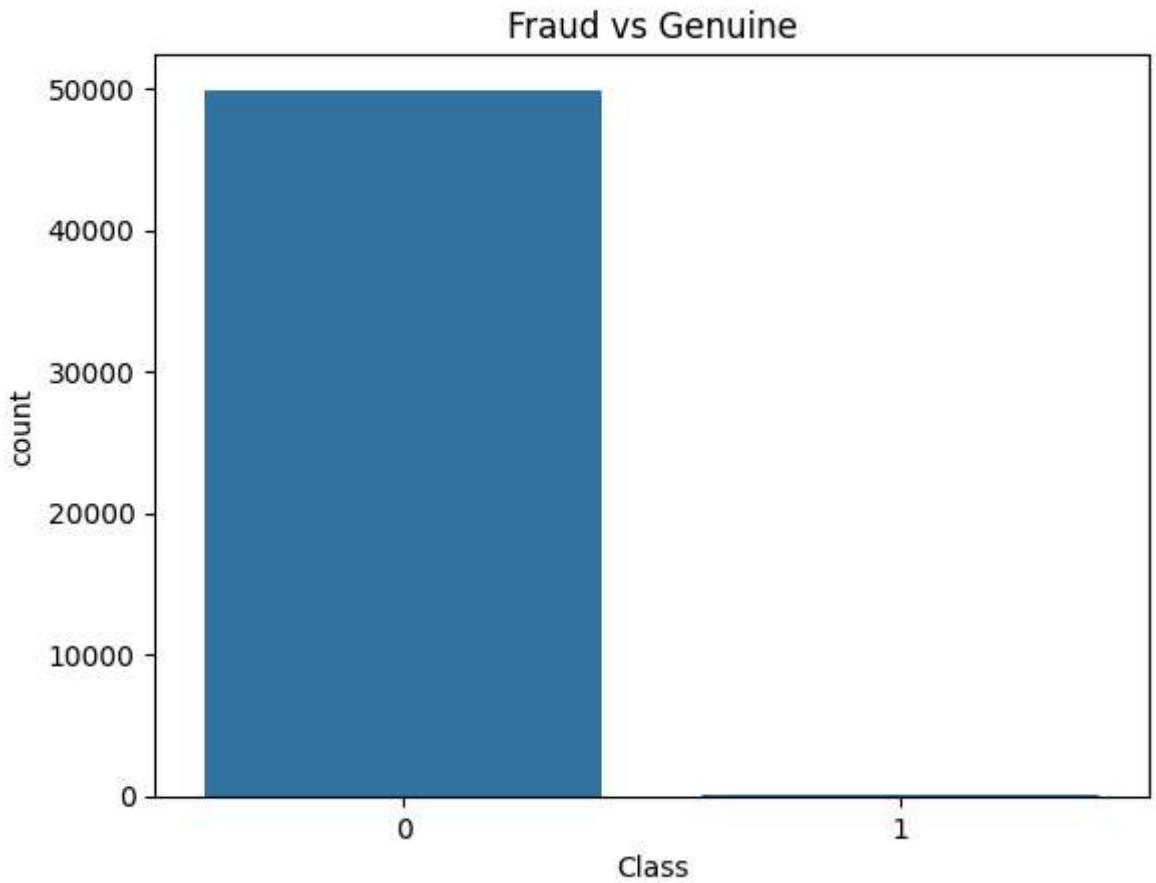
In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.metrics import classification_report, confusion_matrix, precision_reca

from imblearn.over_sampling import SMOTE
```

In [3]:
```
df = pd.read_csv("creditcard.csv")
df.head()

df.info()
df.describe()

df['Class'].value_counts()
df=df.sample(n=50000, random_state=42)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 284807 entries, 0 to 284806
Data columns (total 31 columns):
 #   Column  Non-Null Count   Dtype
---  ------  --------------   -----
 0   Time    284807 non-null  float64
 1   V1      284807 non-null  float64
 2   V2      284807 non-null  float64
 3   V3      284807 non-null  float64
 4   V4      284807 non-null  float64
 5   V5      284807 non-null  float64
 6   V6      284807 non-null  float64
 7   V7      284807 non-null  float64
 8   V8      284807 non-null  float64
 9   V9      284807 non-null  float64
 10  V10     284807 non-null  float64
 11  V11     284807 non-null  float64
 12  V12     284807 non-null  float64
 13  V13     284807 non-null  float64
 14  V14     284807 non-null  float64
 15  V15     284807 non-null  float64
 16  V16     284807 non-null  float64
 17  V17     284807 non-null  float64
 18  V18     284807 non-null  float64
 19  V19     284807 non-null  float64
 20  V20     284807 non-null  float64
 21  V21     284807 non-null  float64
 22  V22     284807 non-null  float64
 23  V23     284807 non-null  float64
 24  V24     284807 non-null  float64
 25  V25     284807 non-null  float64
 26  V26     284807 non-null  float64
 27  V27     284807 non-null  float64
 28  V28     284807 non-null  float64
 29  Amount  284807 non-null  float64
 30  Class   284807 non-null  int64
dtypes: float64(30), int64(1)
memory usage: 67.4 MB
```

In [4]:

```
sns.countplot(x='Class', data=df)
plt.title("Fraud vs Genuine")
plt.show()
```



In [5]:
```
X = df.drop('Class', axis=1)
y = df['Class']

smote = SMOTE(random_state=42)
X_res, y_res = smote.fit_resample(X, y)

print("After SMOTE:")
print(pd.Series(y_res).value_counts())
```

```
After SMOTE:
Class
1    49917
0    49917
Name: count, dtype: int64
```

```
C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\site-packa
ges\joblib\externals\loky\backend\context.py:131: UserWarning: Could not find the nu
mber of physical cores for the following reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by setti
ng LOKY_MAX_CPU_COUNT to the number of cores you want to use.
  warnings.warn(
  File "C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\si
te-packages\joblib\externals\loky\backend\context.py", line 247, in _count_physical_
cores
    cpu_count_physical = _count_physical_cores_win32()
  File "C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\si
te-packages\joblib\externals\loky\backend\context.py", line 299, in _count_physical_
cores_win32
    cpu_info = subprocess.run(
        "wmic CPU Get NumberOfCores /Format:csv".split(),
        capture_output=True,
        text=True,
    )
  File "C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\su
bprocess.py", line 554, in run
    with Popen(*popenargs, **kwargs) as process:
         ~~~~~^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\su
bprocess.py", line 1036, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
    ~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
                        pass_fds, cwd, env,
                        ^^^^^^^^^^^^^^^^^^^
    ...<5 lines>...
                        gid, gids, uid, umask,
                        ^^^^^^^^^^^^^^^^^^^^^^
                        start_new_session, process_group)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\su
bprocess.py", line 1548, in _execute_child
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
                       ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
                             # no special security
                             ^^^^^^^^^^^^^^^^^^^^^
    ...<4 lines>...
                             cwd,
                             ^^^^
                             startupinfo)
                             ^^^^^^^^^^^^
```

In [6]:
```python
X_train, X_test, y_train, y_test = train_test_split(
X_res, y_res, test_size=0.2, random_state=42
)
```

In [7]:
```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

In [8]:
```
lr = LogisticRegression(max_iter=2000)
lr.fit(X_train, y_train)
```

Out[8]:
```
▾ LogisticRegression
▶ Parameters
```

In [11]:
```
rf = RandomForestClassifier(n_estimators=50)
rf.fit(X_train, y_train)
```

Out[11]:
```
▾ RandomForestClassifier
▶ Parameters
```

In [22]:
```
gb = GradientBoostingClassifier(n_estimators=50)
gb.fit(X_train, y_train)
```

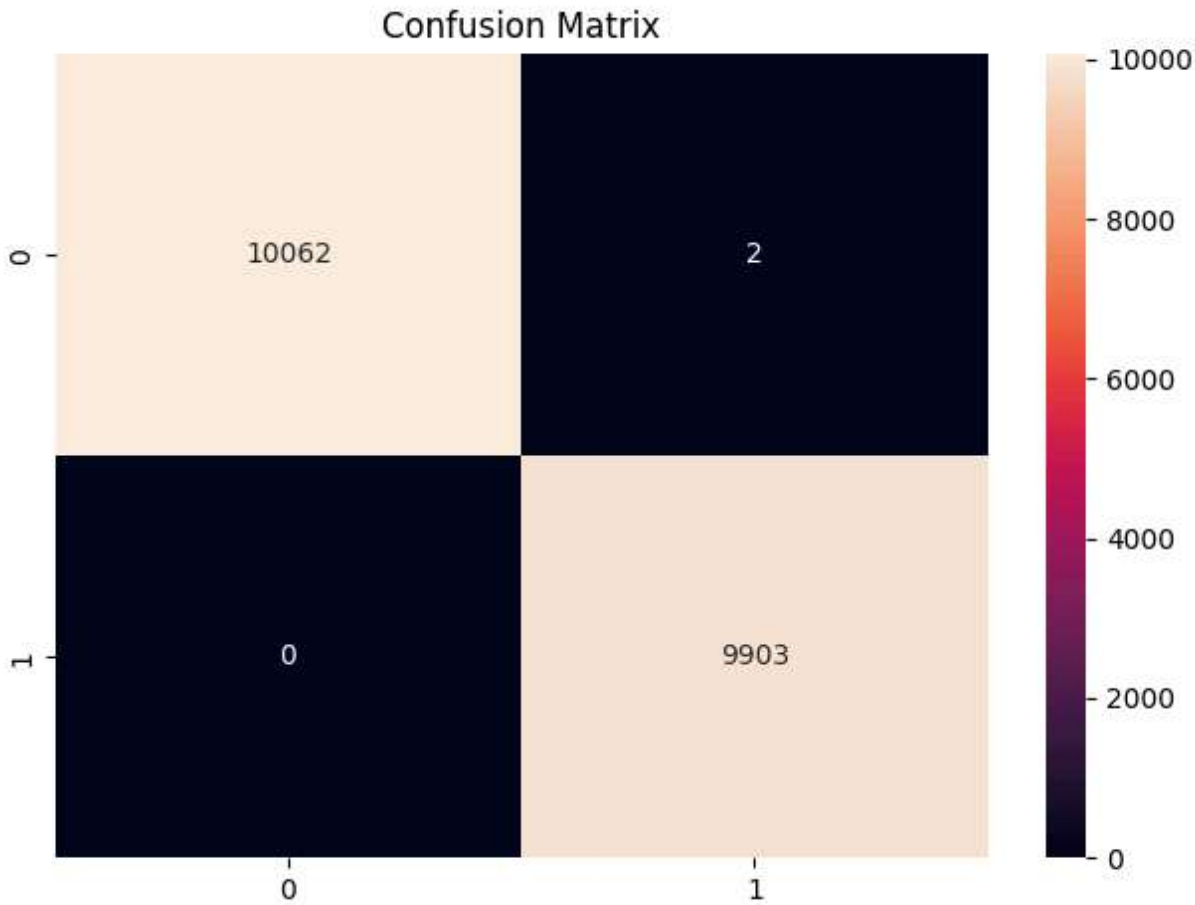
Out[22]:
```
▾ GradientBoostingClassifier
▶ Parameters
```

In [29]:
```
y_pred = rf.predict(X_test)
```

In [24]:
```
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

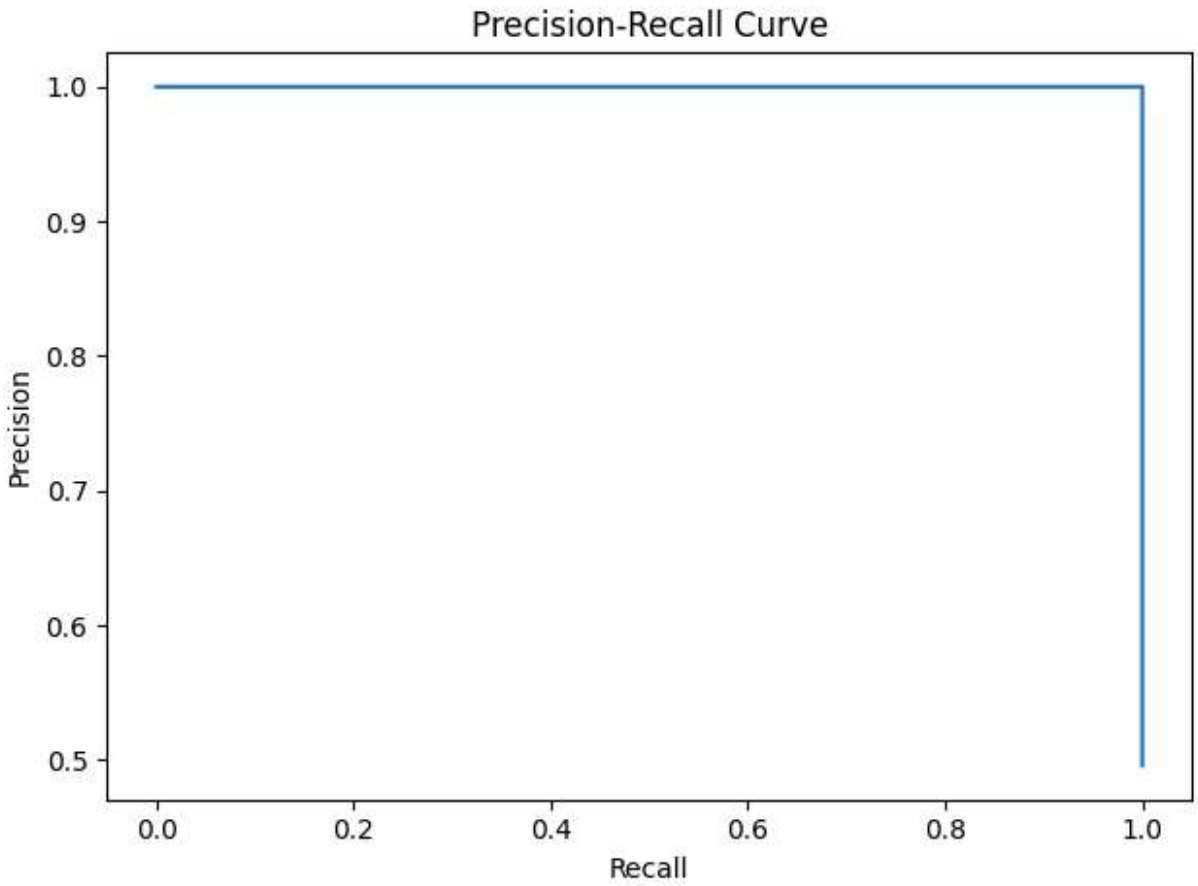
```
Confusion Matrix:
[[10062     2]
 [    0  9903]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     10064
           1       1.00      1.00      1.00      9903

    accuracy                           1.00     19967
   macro avg       1.00      1.00      1.00     19967
weighted avg       1.00      1.00      1.00     19967
```

In [39]:
```
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.tight_layout()
plt.savefig("confusion_matrix.png")
plt.show()
```



In [40]:
```
y_scores = rf.predict_proba(X_test)[:,1]

precision, recall, _ = precision_recall_curve(y_test, y_scores)
```

In [41]:
```
plt.plot(recall, precision)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve")
plt.tight_layout()
plt.savefig("pr_curve.png")
plt.show()
```



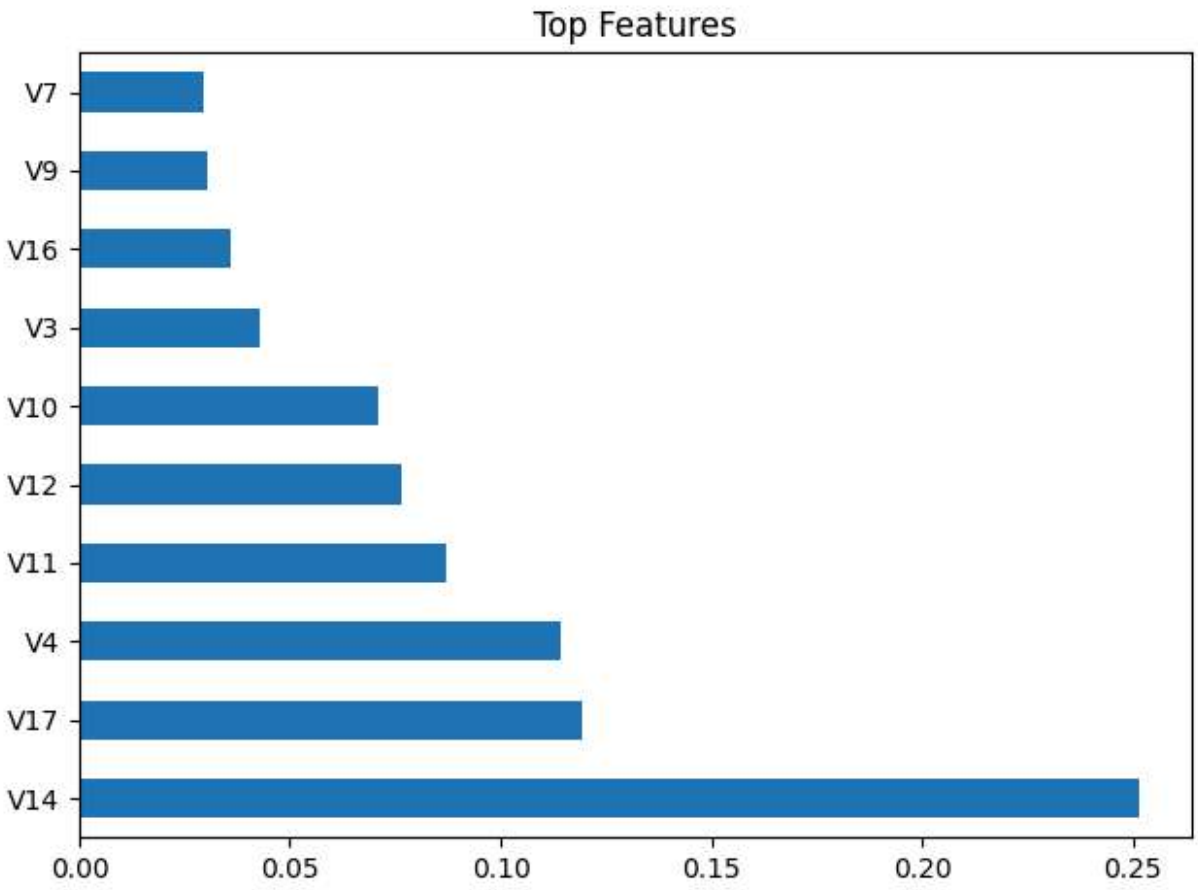
In [42]:
```
df['Risk_Score'] = rf.predict_proba(X)[:,1]
df[['Risk_Score']].head()
```

```
C:\Users\SACHIN RAVICHANDIRAN\AppData\Local\Programs\Python\Python313\Lib\site-packa
ges\sklearn\utils\validation.py:2742: UserWarning: X has feature names, but RandomFo
restClassifier was fitted without feature names
  warnings.warn(
```

Out[42]:

	Risk_Score
43428	0.62
49906	0.20
29474	0.28
276481	0.26
278846	0.52

In [38]:
```
feat_imp=pd.Series(rf.feature_importances_, index=X.columns)
feat_imp.nlargest(10).plot(kind='barh')
plt.title("Top Features")
plt.tight_layout()
plt.savefig("feature_importance.png")
plt.show()
```


Top Features
V7
V9
V16
V3
V10
V12
V11
V4
V17
V14
0.00
0.05
0.10
0.15
0.20
0.25

In [ ]: